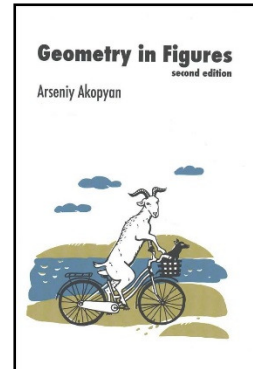
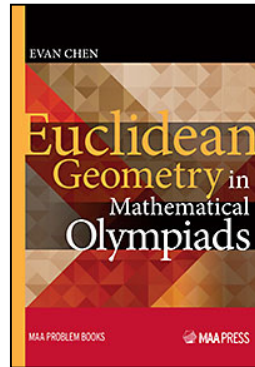
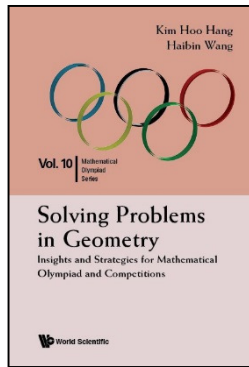


Olympiad Geometry – I

Some reading materials:

- Solving Problems in Geometry
Insights and Strategies for Mathematical Olympiad and Competitions
https://www.worldscientific.com/doi/pdf/10.1142/9789814583756_0001
- Euclidean Geometry in Mathematical Olympiads – Chapter 2 Circles
https://www.maa.org/sites/default/files/pdf/ebooks/pdf/EGMO_chapter2.pdf
- Geometry in Figures
<https://www.mccme.ru/~akopyan/papers/EnGeoFigures.pdf>



0. Notation and Basic Facts

a , b and c are the sides of $\triangle ABC$ opposite to A , B , and C respectively.

$[ABC]$ = area of $\triangle ABC$

$$s = \text{semi-perimeter} = \frac{1}{2}(a + b + c) \quad r = \text{inradius} \quad R = \text{circumradius}$$

Ravi Substitution: $x = s - a$, $y = s - b$, $z = s - c$
i.e. $a = y + z$, $b = x + z$, $c = x + y$.

h_a , h_b , h_c – lengths of the altitudes drawn to the sides a , b , c respectively

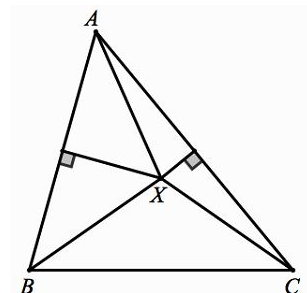
m_a , m_b , m_c – lengths of the medians drawn to the sides a , b , c respectively

i_a , i_b , i_c – lengths of the bisectors of the angles A , B , C respectively

Centres of a triangle

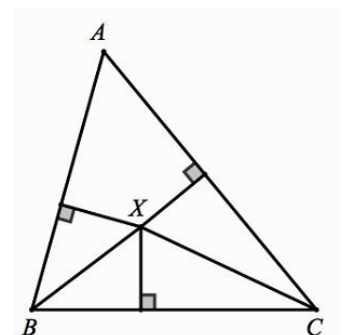
(a) Circumcentre

In a triangle, the perpendicular bisectors of the three sides meet at a point. This point is called the circumcentre of the triangle, usually denoted by O , and it is the centre of the circumcircle of the triangle.



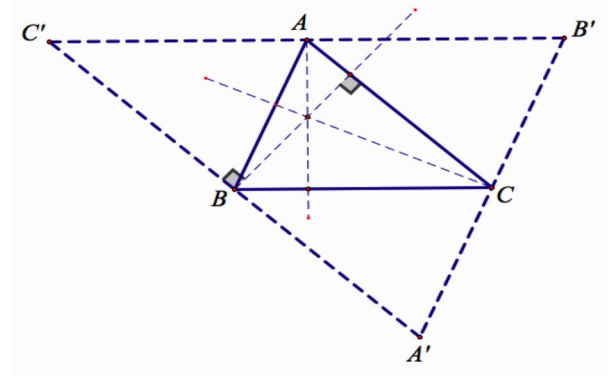
(b) Incentre

In a triangle, the internal angle bisectors meet at a point. This point is called the incentre of the triangle, usually denoted by I , and it is the centre of the incircle of the triangle.

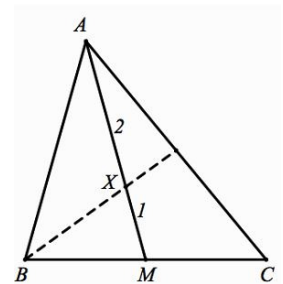


(c) **Orthocentre**

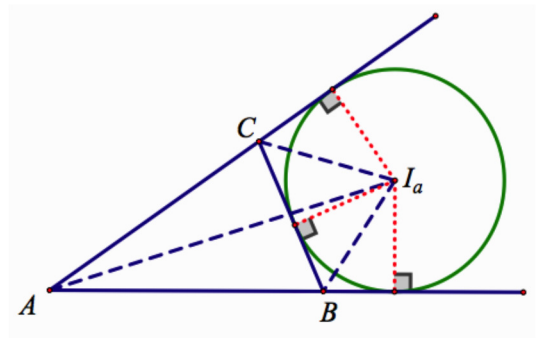
In a triangle, the altitudes meet at a single point. This point is called the orthocentre of the triangle and is usually denoted by H .

(d) **Centroid**

In a triangle, the medians meet at a point. This point is called the centroid of the triangle and is usually denoted by G .

(e) **Excentres**

In triangle ABC the A -angle bisector and the bisectors of external angle B and C meet at a point. This point is called the A -excentre of $\triangle ABC$, usually denoted by I_a and it is the centre of the A -excircle (circle tangent to the side BC and to the extended sidelines AB and AC). Similarly, we define points I_b and I_c .

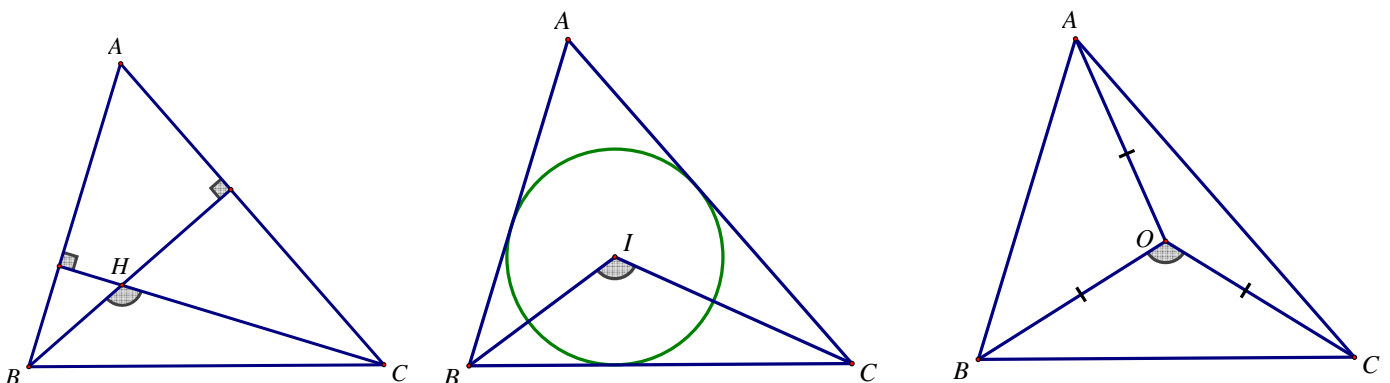
**Angle at centres**

Let ABC be a triangle with orthocentre H , incentre I and circumcentre O . Then:

(a) If triangle ABC is acute, then $\angle BHC = 180^\circ - \angle A$.

(b) $\angle BIC = 90^\circ + \frac{1}{2}\angle A$.

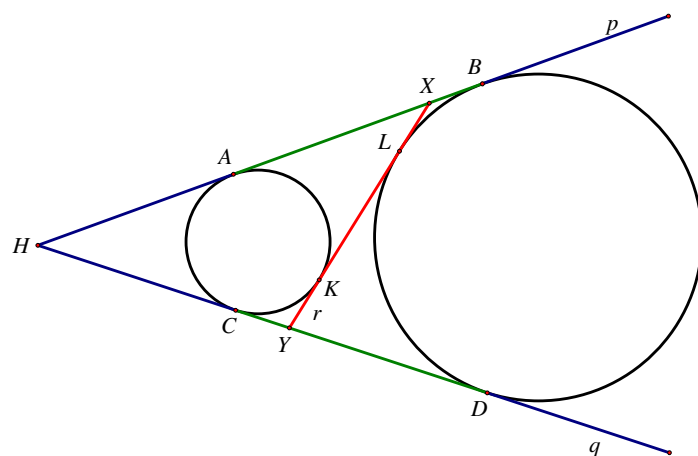
(c) If $\angle A$ is acute, then $\angle BOC = 2\angle A$.



Lengths of tangents

Let p, q be common external tangents of circles ω_1 and ω_2 . Denote by A, B the points of tangency of p with ω_1 and ω_2 respectively, and by C, D the points of tangency of q with ω_1 and ω_2 respectively. Then:

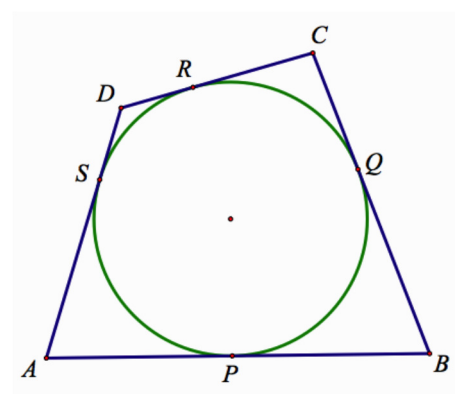
- $AB = CD$.
- If the two circles are nonintersecting and their common internal tangent r intersects p and q at points X, Y , respectively, we have $AB = CD = XY$.



Example 0.1 (Pilot's Theorem) [lengths of tangents]

Let $ABCD$ be a quadrilateral with an inscribed circle.

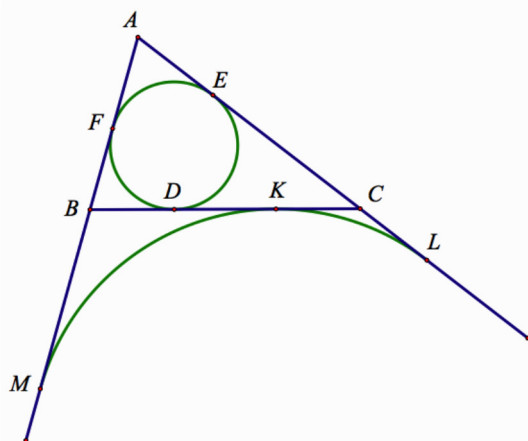
Then $AB + CD = BC + DA$.



Segments of triangle related to tangency

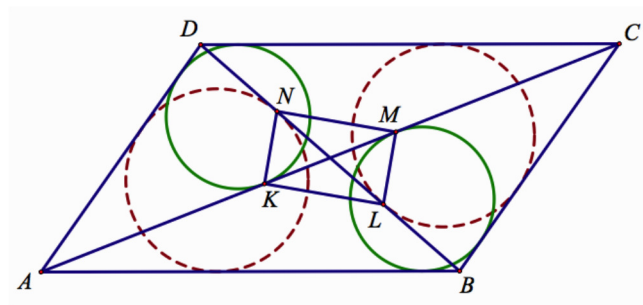
Let ABC be a triangle with semiperimeter s . Denote by D, E, F the points of tangency of the incircle with sides BC, CA, AB respectively. Also let the A -excircle touch the lines BC, CA, AB at points K, L, M respectively. Then the following holds:

- $AE = AF = s - a$; $BD = BF = s - b$; $CD = CE = s - c$
- $AL = AM = s$; hence $CL = CK = s - b$ and $BK = BM = s - c$
- Points K and D are symmetric with respect to the midpoint of BC .

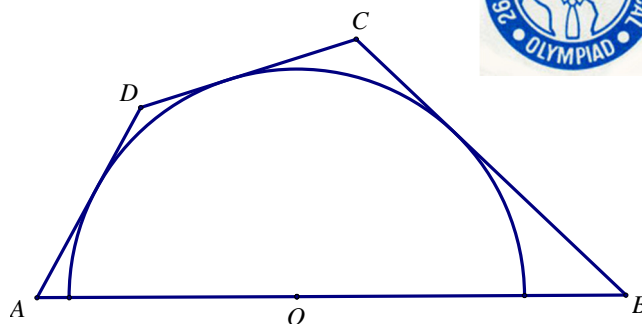


Example 0.2 [segments lengths related to tangency]

Let $ABCD$ be a parallelogram with $AB > BC$. Let K, M be the points of tangency of the incircles of triangles ACD and ABC with AC , respectively. Similarly, let L, N be the points of tangency of the incircles of triangles BCD and ABD with BD respectively. Prove that $KLMN$ is a rectangle.

**Example 0.3 (IMO 1985 P1)**

A circle has centre on the side AB of the cyclic quadrilateral $ABCD$. The other three sides are tangent to the circle. Prove that $AD + BC = AB$.



1. Metric Relations

(Extended) Sine Formula: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

Cosine Formula: $a^2 = b^2 + c^2 - 2bc \cos A$

Areas of triangle

Let ABC be a triangle. Then we can calculate its area in the following ways:

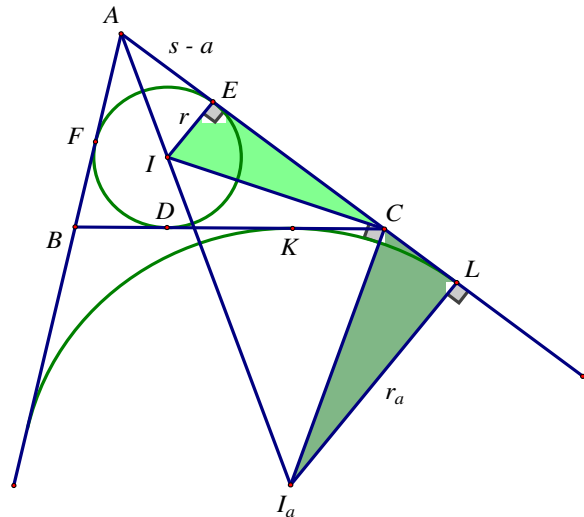
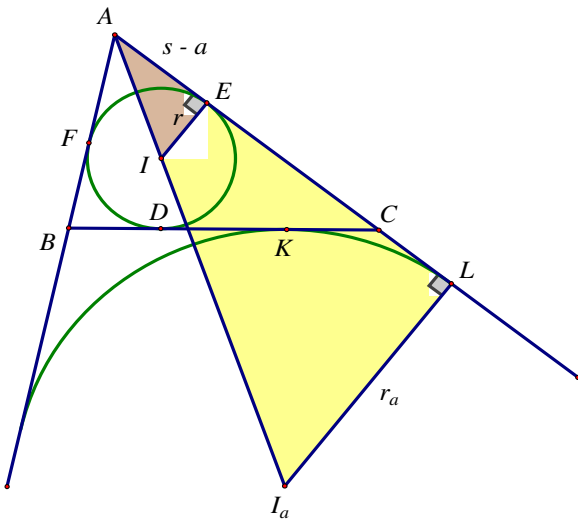
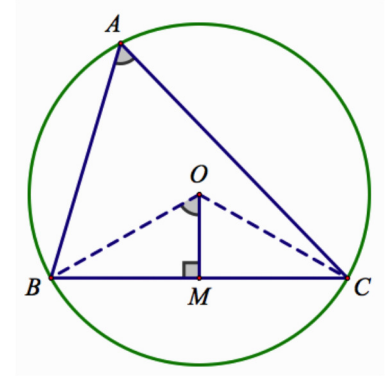
(a) $[ABC] = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B = \frac{abc}{4R}$

(b) $[ABC] = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{xyz}$ (Heron's Formula)

(c) $[ABC] = sr = r_a(s-a) = r_b(s-b) = r_c(s-c)$

(d) Let M be a point on the line BC and AM makes an angle θ (either acute or obtuse) with BC , then

$$[ABC] = \frac{1}{2}AM \times BC \times \sin \theta.$$



xyz formula

In triangle ABC , we can find the area $[ABC]$, inradius r , and circumradius R in terms of xyz as follows:

(a) $[ABC] = \sqrt{(x+y+z)xyz}$

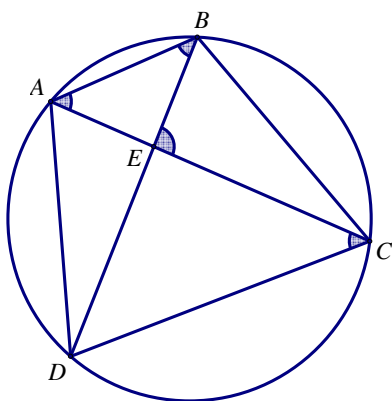
(b) $r = \sqrt{\frac{xyz}{x+y+z}}$

(c) $R = \frac{(y+z)(z+x)(x+y)}{4\sqrt{xyz(x+y+z)}}$

Ptolemy's Theorem

If the convex quadrilateral $ABCD$ is inscribed in a circle, then

$$AB \times CD + AD \times BC = AC \times BD.$$

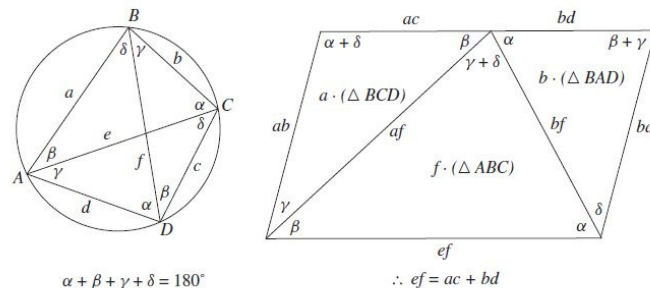


Proof:

Proof Without Words: Ptolemy's Theorem

William Derrick (DerrickWR@mso.umt.edu) and
James Hirstein (hirstein@mso.umt.edu)

In an inscribed quadrilateral, the product of the diagonals is equal to the sum of the products of the opposite sides.



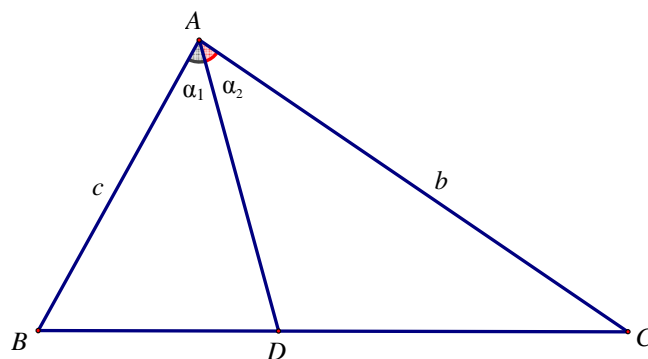
Summary. A visual proof of Ptolemy's theorem.

<http://dx.doi.org/10.4169/college.math.j.43.5.386>
MSC: 97G20

Ratio Lemma

In triangle ABC , let $D \in BC$ and denote by α_1 and α_2 the angles BAD and DAC respectively. Then

$$\frac{BD}{DC} = \frac{AB \sin \alpha_1}{AC \sin \alpha_2}.$$



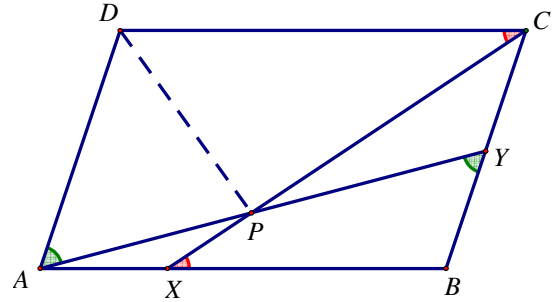
Angle Bisector Theorem

In triangle ABC let AD , $D \in BC$, be the internal angle bisector. Then

$$\frac{BD}{CD} = \frac{c}{b}, \quad BD = \frac{ac}{b+c}, \quad CD = \frac{ab}{b+c}$$

Example 1.1 (Germany 2003)

Let $ABCD$ be a parallelogram. Let X and Y be points on the sides AB and BC respectively, such that $AX = CY$. Prove that the intersection of lines AY and CX lies on the angle bisector of $\angle ADC$.

**Example 1.2 (Criterion for orthogonality)**

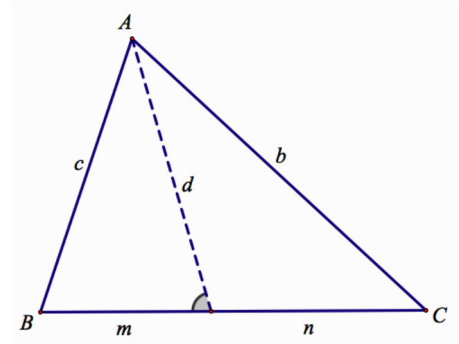
Let AC and BD be two lines in plane. Then $AC \perp BD$ if and only if $AB^2 + CD^2 = AD^2 + BC^2$.

Stewart's Theorem

In triangle ABC let D lie on the side BC . Denote the distance BD , DC and AD by m , n and d respectively.

$$\text{Then } d^2 + mn = b^2 \left(\frac{m}{m+n} \right) + c^2 \left(\frac{n}{m+n} \right)$$

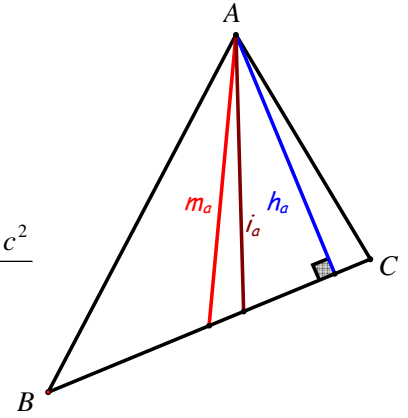
Proof:



Exercises:

1. Prove that $h_a = \frac{bc}{2R}$, $h_b = \frac{ac}{2R}$, $h_c = \frac{ab}{2R}$.

2. Prove that $m_a^2 = \frac{2(b^2 + c^2) - a^2}{4}$, $m_b^2 = \frac{2(a^2 + c^2) - b^2}{4}$, $m_c^2 = \frac{2(a^2 + b^2) - c^2}{4}$



3. Prove that $i_a^2 = bc \frac{[(b+c)^2 - a^2]}{(b+c)^2} = 4bc \frac{s(s-a)}{(b+c)^2}$ $i_b^2 = ca \frac{[(c+a)^2 - b^2]}{(c+a)^2} = 4ca \frac{s(s-b)}{(c+a)^2}$
 $i_c^2 = ab \frac{[(a+b)^2 - c^2]}{(a+b)^2} = 4ab \frac{s(s-c)}{(a+b)^2}$.

4. **(IMO Preliminary Contest 2021 Problem 20)**

In $\triangle ABC$, $AB = 15$ and $AC = 13$. $DEFG$ is a square such that D is on AC , F is on AB and G is the mid-point of BC . If $AD = 11$ and $AF > FB$, find the area of $DEFG$.